

DNS Wireshark Lab | Question 1

What is the destination port for the DNS query message? What is the source port of DNS response message?

Answer: The destination port of the DNS query is 53 (packet 162). The source port of the DNS response is also 53 (packet 163).

Query packet 162: UDP Destination Port: 53.

The image shows a Wireshark packet capture window with the 'dns' filter applied. The packet list shows two packets: packet 162 (DNS Standard query) and packet 163 (DNS Standard query response). Packet 162 is selected, and its details pane shows the following information:

- User Datagram Protocol, Src Port: 63868, Dst Port: 53
 - Source Port: 63868
 - Destination Port: 53
 - Length: 34
 - Checksum: 0xb3c2 [unverified]
 - [Checksum Status: Unverified]
 - [Stream index: 7]
 - [Stream Packet Number: 1]
 - [Timestamps]
 - UDP payload (26 bytes)
- Domain Name System (query)
 - Transaction ID: 0x0004
 - Flags: 0x0100 Standard query
 - Questions: 1
 - Answer RRs: 0
 - Authority RRs: 0
 - Additional RRs: 0

DNS Wireshark Lab | Question 1 (continued)

What is the destination port for the DNS query message? What is the source port of DNS response message?

Answer: The destination port of the DNS query is 53 (packet 162). The source port of the DNS response is also 53 (packet 163).

Response packet 163: UDP Source Port: 53.

The image shows a Wireshark packet capture window with the filter 'dns'. The packet list shows two packets:

No.	Time	Source	Destination	Protocol	Length	Info
162	15.732646600			DNS	68	Standard query 0x0004 A cuny.edu
163	15.734858900			DNS	84	Standard query response 0x0004 A cuny.edu A 128.228.254.200

The packet details pane for packet 163 (highlighted) shows the following structure:

- User Datagram Protocol, Src Port: 53, Dst Port: 63868
 - Source Port: 53
 - Destination Port: 63868
 - Length: 50
 - Checksum: 0xf23f [unverified]
 - [Checksum Status: Unverified]
 - [Stream index: 7]
 - [Stream Packet Number: 2]
 - [Timestamps]
 - UDP payload (42 bytes)
- Domain Name System (response)
 - Transaction ID: 0x0004
 - Flags: 0x8180 Standard query response, No error
 - Questions: 1
 - Answer RRs: 1
 - Authority RRs: 0
 - Additional RRs: 0

DNS Wireshark Lab | Question 2

To what IP address is the DNS query message sent?

Answer: The DNS query is sent to [REDACTED], my home router/DNS server.

Query packet 162: IPv4 Destination Address: [REDACTED].

No.	Time	Source	Destination	Protocol	Length	Info
162	15.732646600			DNS	68	Standard query 0x0004 A cuny.edu
163	15.734858900			DNS	84	Standard query response 0x0004 A cuny.edu A 128.228.254.200


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0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 54
  Identification: 0x14f7 (5367)
> 000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: UDP (17)
Header Checksum: 0xa1cb [validation disabled]
[Header checksum status: Unverified]
Source Address: [REDACTED]
Destination Address: [REDACTED]
[Stream index: 1]
> User Datagram Protocol, Src Port: 63868, Dst Port: 53
> Domain Name System (query)
```

DNS Wireshark Lab | Question 3

Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

Answer: The query is Type A (1), requesting the IPv4 address of cuny.edu. It contains no answers, as shown by Answer RRs: 0.

Query packet 162: Type A (1) and Answer RRs: 0.

The image shows a Wireshark packet capture of a DNS query. The packet list pane at the top shows two packets: packet 162 (15.732646600) and packet 163 (15.734858900). Packet 162 is a DNS Standard query for cuny.edu, and packet 163 is the corresponding Standard query response. The packet details pane for packet 162 is expanded, showing the following structure:

- User Datagram Protocol, Src Port: 63868, Dst Port: 53
- Domain Name System (query)
 - Transaction ID: 0x0004
 - Flags: 0x0100 Standard query
 - Questions: 1
 - Answer RRs: 0
 - Authority RRs: 0
 - Additional RRs: 0
 - Queries
 - cuny.edu: type A, class IN
 - Name: cuny.edu
 - [Name Length: 8]
 - [Label Count: 2]
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)

The packet bytes pane at the bottom shows the raw data of the packet, with a search bar and a search button.

DNS Wireshark Lab | Question 4

Examine the DNS response message. How many “answers” are provided? What do each of these answers contain?

Answer: The response provides one answer (Answer RRs: 1): an A record mapping cuny.edu to 128.228.254.200. The record has class IN, a TTL of 289 seconds (4 minutes, 49 seconds), and a data length of 4 bytes.

Response packet 163: Answer RRs: 1 and the expanded A record.

The image shows a Wireshark network packet capture. The top pane displays a list of packets. Packet 163, at time 15.734858900, is a DNS Standard query response from 15.732646600 to 15.734858900. The bottom pane shows the expanded details of packet 163. The 'Answers' section is expanded, showing a single A record for cuny.edu with IP address 128.228.254.200. The 'Request In: 162' is also visible.

Time	Source	Destination	Protocol	Length	Info
162 15.732646600			DNS	68	Standard query 0x0004 A cuny.edu
163 15.734858900			DNS	84	Standard query response 0x0004 A cuny.edu A 128.228.254.200

```
.... 1... = Recursion available: Server can do recursive queries
.... 0... = Z: reserved (0)
.... ..0. = Answer authenticated: Answer/authority portion was not authenticated by the server
.... ..0. = Non-authenticated data: Unacceptable
.... ..0000 = Reply code: No error (0)

Questions: 1
Answer RRs: 1
Authority RRs: 0
Additional RRs: 0
  Queries
  Answers
    cuny.edu: type A, class IN, addr 128.228.254.200
      Name: cuny.edu
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 289 (4 minutes, 49 seconds)
      Data length: 4
      Address: 128.228.254.200
      [Request In: 162]
      [Time: 2.212300 milliseconds]
```

DNS Wireshark Lab | Question 5

Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

Answer: The query is Type NS (2), requesting the nameservers for cuny.edu. It contains no answers, as shown by Answer RRs: 0.

Query packet 123: Type NS (2) and Answer RRs: 0.

The image shows a Wireshark network packet capture. The top pane displays a list of packets. Packet 123 is a DNS query from 13.624903600 to 13.633197200. Packet 124 is the corresponding DNS response from 13.633197200 to 13.624903600. The bottom pane shows the details of packet 123, which is a Domain Name System (query) with Transaction ID 0x0003. The query is for the domain cuny.edu, type NS, class IN. The response (packet 124) is a Standard query response with Transaction ID 0x0003, containing the nameservers for cuny.edu: harlem.cis.cuny.edu, ns4.level3.net, and hudson.cis.cuny.edu.

No.	Time	Source	Destination	Protocol	Length	Info
123	13.624903600			DNS	68	Standard query 0x0003 NS cuny.edu
124	13.633197200			DNS	372	Standard query response 0x0003 NS cuny.edu NS harlem.cis.cuny.edu NS ns4.level3.net NS hudson.cis.cuny.edu

Domain Name System (query)
Transaction ID: 0x0003
Flags: 0x0100 Standard query
0... .. = Response: Message is a query
.000 0... .. = Opcode: Standard query (0)
... ..0. = Truncated: Message is not truncated
... ..1 = Recursion desired: Do query recursively
... ..0.. = Z: reserved (0)
... ..0 = Non-authenticated data: Unacceptable
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
cuny.edu: type NS, class IN
Name: cuny.edu
[Name Length: 8]
[Label Count: 2]
Type: NS (2) (authoritative Name Server)
Class: IN (0x0001)
[Response In: 124]

DNS Wireshark Lab | Question 6

Examine the DNS response message. What CUNY nameservers does the response message provide?

Answer: The response provides seven nameservers (Answer RRs: 7):
harlem.cis.cuny.edu; ns4.level3.net; hudson.cis.cuny.edu; acme.ucc.cuny.edu;
ns3.level3.net; lavinia.cis.cuny.edu; ext-ns1.columbia.edu.

Response packet 124: Answer RRs: 7 and all seven NS records.

The image shows a Wireshark packet capture of a DNS response. The packet list on the left shows packet 124 at time 13.633197200. The packet details pane on the right shows the DNS response structure. The packet bytes pane at the bottom shows the raw data and its interpretation.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
123				DNS	68	Standard query 0x0003 NS cuny.edu
124	13.633197200			DNS	372	Standard query response 0x0003 NS cuny.edu NS harlem.cis.cuny.edu NS ns4.le

Packet Details:

- DNS
- Standard query response 0x0003 NS cuny.edu NS harlem.cis.cuny.edu NS ns4.le

Packet Bytes:

```
.... 0.. .... = Authoritative: Server is not an authority for domain
.... 0.. .... = Truncated: Message is not truncated
.... 1.. .... = Recursion desired: Do query recursively
.... 1.. .... = Recursion available: Server can do recursive queries
.... 0.. .... = Z: reserved (0)
.... 0.. .... = Answer authenticated: Answer/authority portion was not authenticated by the serv
.... 0.. .... = Non-authenticated data: Unacceptable
.... 0000 = Reply code: No error (0)

Questions: 1
Answer RRs: 7
Authority RRs: 0
Additional RRs: 7
Queries
Answers
  cuny.edu: type NS, class IN, ns harlem.cis.cuny.edu
  cuny.edu: type NS, class IN, ns ns4.level3.net
  cuny.edu: type NS, class IN, ns hudson.cis.cuny.edu
  cuny.edu: type NS, class IN, ns acme.ucc.cuny.edu
  cuny.edu: type NS, class IN, ns ns3.level3.net
  cuny.edu: type NS, class IN, ns lavinia.cis.cuny.edu
  cuny.edu: type NS, class IN, ns ext-ns1.columbia.edu
Additional records
[Request In: 123]
[Time: 8.293600 milliseconds]
```

DNS Wireshark Lab | Question 7

Does this response message also provide the IP addresses of the CUNY nameservers?

Answer: Yes. The Additional records section provides addresses for five of the seven nameservers. Its seven records include five A (IPv4) records and two AAAA (IPv6) records. No addresses are included for harlem.cis.cuny.edu or hudson.cis.cuny.edu.

Nameserver	IP address(es)
lavinia.cis.cuny.edu	128.228.5.64
ext-ns1.columbia.edu	128.59.1.1; 2001:18d8:1::1:1; 2001:18d8::35
ns4.level3.net	64.132.94.251
acme.ucc.cuny.edu	128.228.1.10
ns3.level3.net	216.136.95.12

Response packet 124: Additional RRs: 7 and the address records.

The image shows a Wireshark packet capture of a DNS response. The packet list pane shows packet 124, a 'Standard query response' from 13.633197200. The packet details pane shows the 'Additional records' section expanded, listing seven records: five A records and two AAAA records for various CUNY nameservers. The packet bytes pane shows the raw DNS data.

Packet 124: Standard query response 0x0003 NS cuny.edu NS harlem.cis.cuny.edu NS ns4.le

Transaction ID: 0x0003

Flags: 0x8180 Standard query response, No error

Questions: 1

Answer RRs: 7

Authority RRs: 0

Additional RRs: 7

Queries

Answers

Additional records

- lavinia.cis.cuny.edu: type A, class IN, addr 128.228.5.64
- ext-ns1.columbia.edu: type A, class IN, addr 128.59.1.1
- ext-ns1.columbia.edu: type AAAA, class IN, addr 2001:18d8:1::1:1
- ext-ns1.columbia.edu: type AAAA, class IN, addr 2001:18d8::35
- ns4.level3.net: type A, class IN, addr 64.132.94.251
- acme.ucc.cuny.edu: type A, class IN, addr 128.228.1.10
- ns3.level3.net: type A, class IN, addr 216.136.95.12

[Request In: 123]

[Time: 8.293600 milliseconds]

DNS Wireshark Lab | Question 8

To what IP address is the DNS query message to resolve cuny.edu sent to? Is this IP address same as in Q2. If not, what does the IP address correspond to?

Answer: The query is sent to 128.228.1.10, the IP address of acme.ucc.cuny.edu, CUNY's authoritative DNS server. This is different from Q2's [REDACTED], which is my home router/DNS server.

Query packet 108: IPv4 Destination Address: 128.228.1.10.

The image shows a Wireshark packet capture window with the 'dns' filter applied. The packet list shows two packets:

No.	Time	Source	Destination	Protocol	Length	Info
108	6.772395600			DNS	68	Standard query 0x0004 A cuny.edu
109	7.376536400			DNS	84	Standard query response 0x0004 A cuny.edu A 128.228.254.

The packet details pane for packet 108 (the query) is expanded, showing the following fields:

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 54
- Identification: 0x1b57 (6999)
- 0000. = Flags: 0x0
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: UDP (17)
- Header Checksum: 0xdb26 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: [REDACTED]
- Destination Address: 128.228.1.10
- [Stream index: 3]
- User Datagram Protocol, Src Port: 53089, Dst Port: 53
- Domain Name System (query)

DNS Wireshark Lab | Question 9

Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

Answer: The query is Type A (1), requesting the IPv4 address of cuny.edu. It contains no answers, as shown by Answer RRs: 0.

Query packet 108: Type A (1) and Answer RRs: 0.

The image shows a Wireshark network packet capture. The top pane displays a list of packets. Packet 108 is a DNS query (Type A) for cuny.edu, and packet 109 is the corresponding DNS response. The bottom pane shows the detailed view of packet 108, which is a User Datagram Protocol (UDP) packet from source port 53089 to destination port 53. The DNS section is expanded, showing a standard query for cuny.edu with transaction ID 0x0004. The query is for Type A (Host Address) with class IN. The response in packet 109 is also visible in the top pane, showing a standard query response for cuny.edu with the IP address 128.228.254.

No.	Time	Source	Destination	Protocol	Length	Info
108	6.772395600			DNS	68	Standard query 0x0004 A cuny.edu
109	7.376536400			DNS	84	Standard query response 0x0004 A cuny.edu A 128.228.254.

Packet 108 details:

- User Datagram Protocol, Src Port: 53089, Dst Port: 53
- Domain Name System (query)
 - Transaction ID: 0x0004
 - Flags: 0x0100 Standard query
 - Questions: 1
 - Answer RRs: 0
 - Authority RRs: 0
 - Additional RRs: 0
- Queries
 - cuny.edu: type A, class IN
 - Name: cuny.edu
 - [Name Length: 8]
 - [Label Count: 2]
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)
 - [Response In: 109]

DNS Wireshark Lab | Question 10

Examine the DNS response message. How many “answers” are provided? What does each of these answers contain?

Answer: The response provides one answer (Answer RRs: 1): an A record mapping cuny.edu to 128.228.254.200. The record has class IN, a TTL of 600 seconds (10 minutes), and a data length of 4 bytes.

Response packet 109: Answer RRs: 1 and the expanded A record.

The image shows a Wireshark network traffic capture. The top pane displays a list of packets. Packet 109 is selected, showing it is a DNS Standard query response from 6.194.244.200 to 6.772.395.600. The bottom pane shows the expanded details of packet 109, which is a Domain Name System (response) message. The details include:

- Transaction ID: 0x0004
- Flags: 0x8500 Standard query response, No error
- Questions: 1
- Answer RRs: 1
- Authority RRs: 0
- Additional RRs: 0
- Queries
- Answers
 - cuny.edu: type A, class IN, addr 128.228.254.200

The expanded A record details are shown below the list:

- Name: cuny.edu
- Type: A (1) (Host Address)
- Class: IN (0x0001)
- Time to live: 600 (10 minutes)
- Data length: 4
- Address: 128.228.254.200

At the bottom, it indicates the request was in packet 108 and the time taken was 604.140800 milliseconds.